
House Price Prediction — Ames Housing Dataset

1. Introduction

This project predicts residential house sale prices in Ames, Iowa using the [Kaggle Ames Housing Dataset](#). Only numerical features were used — square footage, room counts, lot size, year built — to focus on preprocessing, feature engineering, and regression modeling. The core algorithm was **Linear Regression**.

2. Dataset

Files: train.csv, test.csv | **Target:** SalePrice (USD)

After filtering to numerical columns only, key features included: OverallQual, GrLivArea, GarageCars, GarageArea, TotalBsmtSF, YearBuilt, FullBath, TotRmsAbvGrd, LotArea.

3. Exploratory Data Analysis

Top correlates with SalePrice: OverallQual > GrLivArea > GarageCars > GarageArea > TotalBsmtSF

Key findings:

- Larger living areas and better build quality strongly predict higher prices.
 - Garage and basement size also correlate positively with sale price.
 - SalePrice was right-skewed → applied **$\log(1 + \text{SalePrice})$** transformation to normalize.
 - Outliers (oversized homes at low prices, large garages) were retained as genuine observations.
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4. Data Cleaning

Issue	Treatment
Continuous missing values (e.g. LotFrontage)	Median imputation
Structural absences (e.g. GarageArea, GarageCars)	Zero imputation
Highly correlated pairs ($r > 0.80$)	Reviewed; some retained
Feature scale variance	StandardScaler applied

5. Feature Engineering

Feature	Formula
TotalSF	$1stFlrSF + 2ndFlrSF + TotalBsmtSF$
HouseAge	$YrSold - YearBuilt$
TotalBathrooms	$FullBath + 0.5 \times HalfBath + BsmtFullBath + 0.5 \times BsmtHalfBath$
YearsSinceRemodel	$YrSold - YearRemodAdd$

6. Model & Results

Pipeline: StandardScaler → Linear Regression | **Validation:** 5-fold cross-validation | **Metric:** RMSE on $\log(\text{SalePrice})$

Result: $RMSE < 0.16$ — meeting the Kaggle competition benchmark.

7. Limitations & Next Steps

The model struggled with luxury homes and unusual properties not well-represented by numerical features alone.

Potential improvements: add categorical features · Ridge/Lasso regularization · Random Forest or XGBoost · deeper feature engineering.

A Linear Regression pipeline using only numerical features achieved strong predictive performance on the Ames Housing Dataset. The workflow — cleaning, EDA, log-transformation, feature engineering, scaling, and cross-validation — demonstrates that a well-prepared simple model can be highly competitive.

The key changes made: merged thin sections, converted prose lists to tables, cut repetitive explanations, tightened all phrasing, and reorganized results into a scannable summary. The content is preserved but roughly 60% shorter.